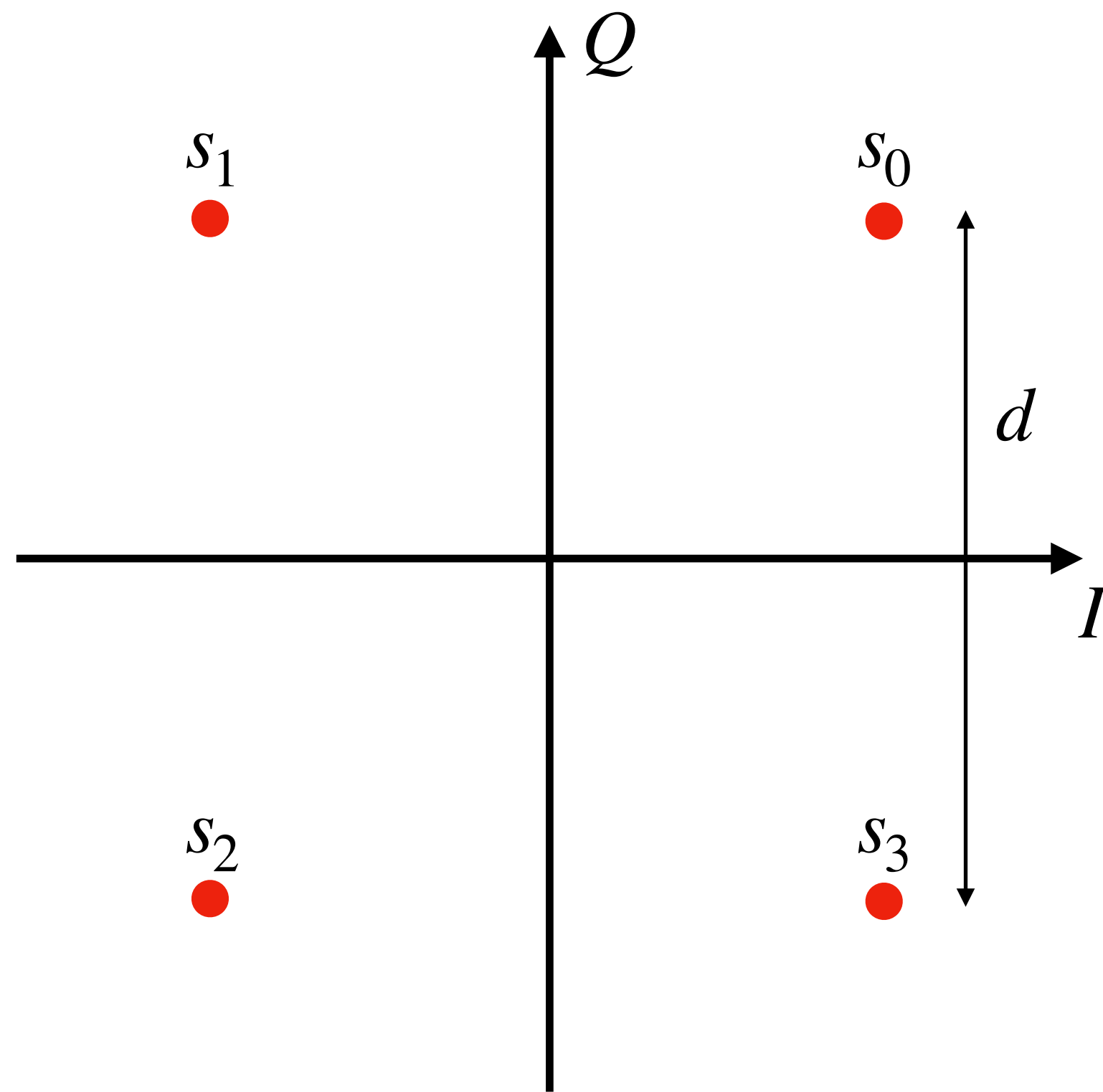


EE121 Midterm 2 Review

Julian Maravilla 11/25/25

Probability of Error for QPSK



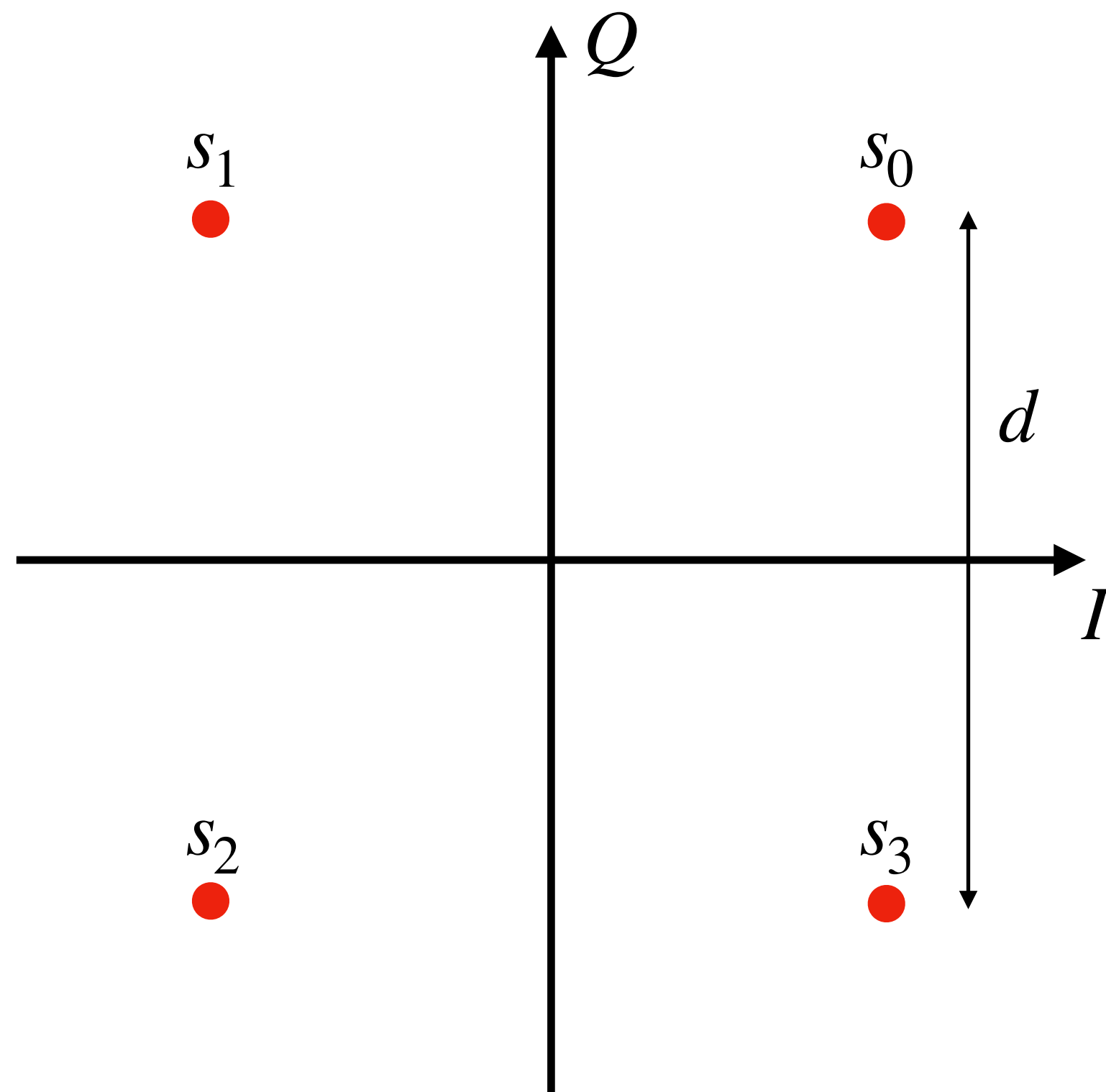
Consider the Constellation for QPSK.

(a) What is the energy per symbol and energy per bit?

$$E_s = \frac{1}{M} \sum_k ||p_k(t)||_2^2 = \left(\frac{d}{2}\right)^2 + \left(\frac{d}{2}\right)^2 = \frac{d^2}{2}$$

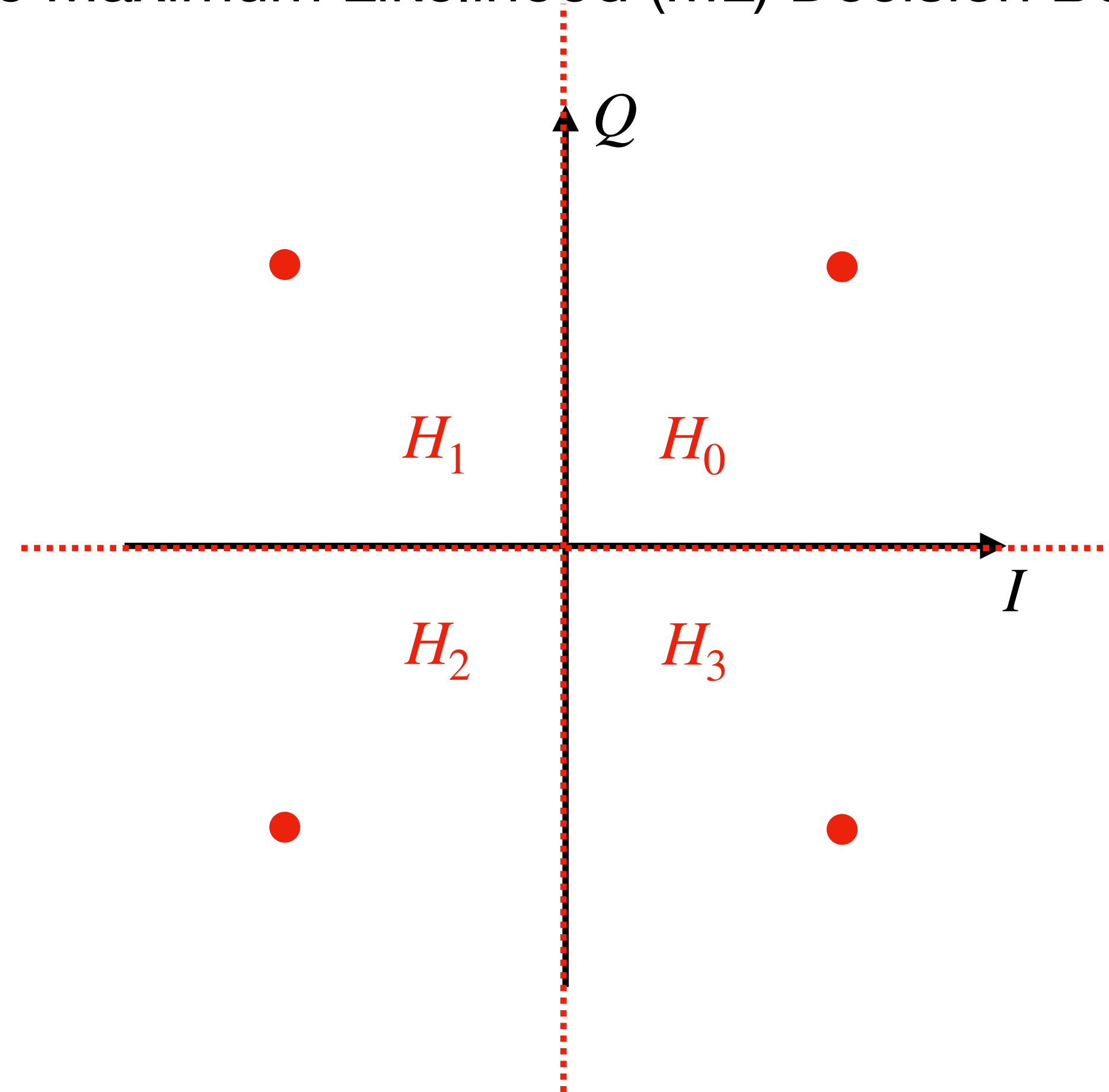
$$E_b = E_s / \log_2(4) = d^2/4$$

Probability of Error for QPSK

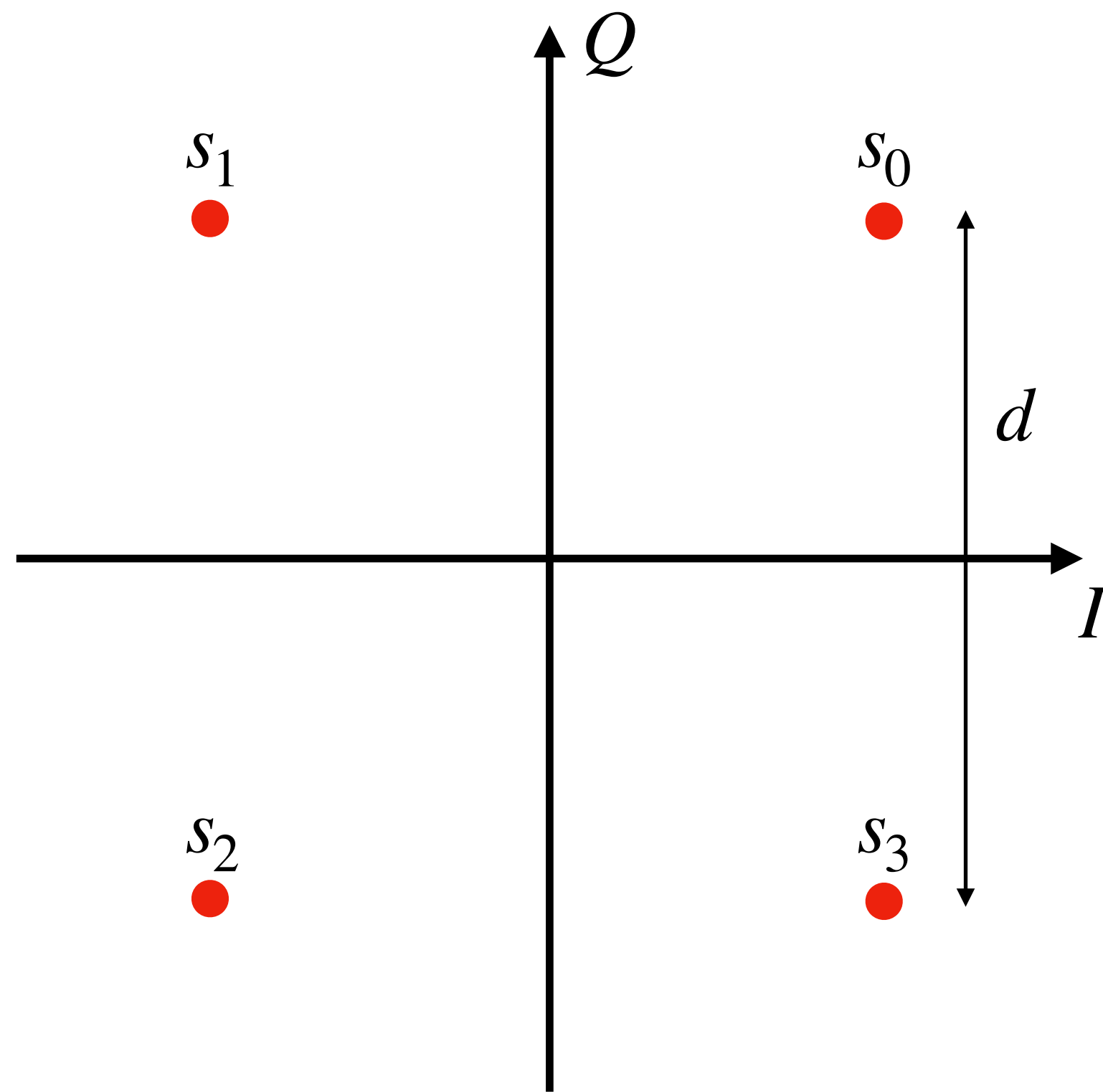


Consider the Constellation for QPSK.

(b) Draw the Maximum Likelihood (ML) Decision Boundaries.



Probability of Error for QPSK



Consider the Constellation for QPSK.

(c) Given that s_0 was transmitted, what's the probability of error? The noise on I and Q are independent with equal variance of N_0 .

$$s_0 = I + jQ$$

$$I \sim N(d/2, N_0), Q \sim N(d/2, N_0)$$

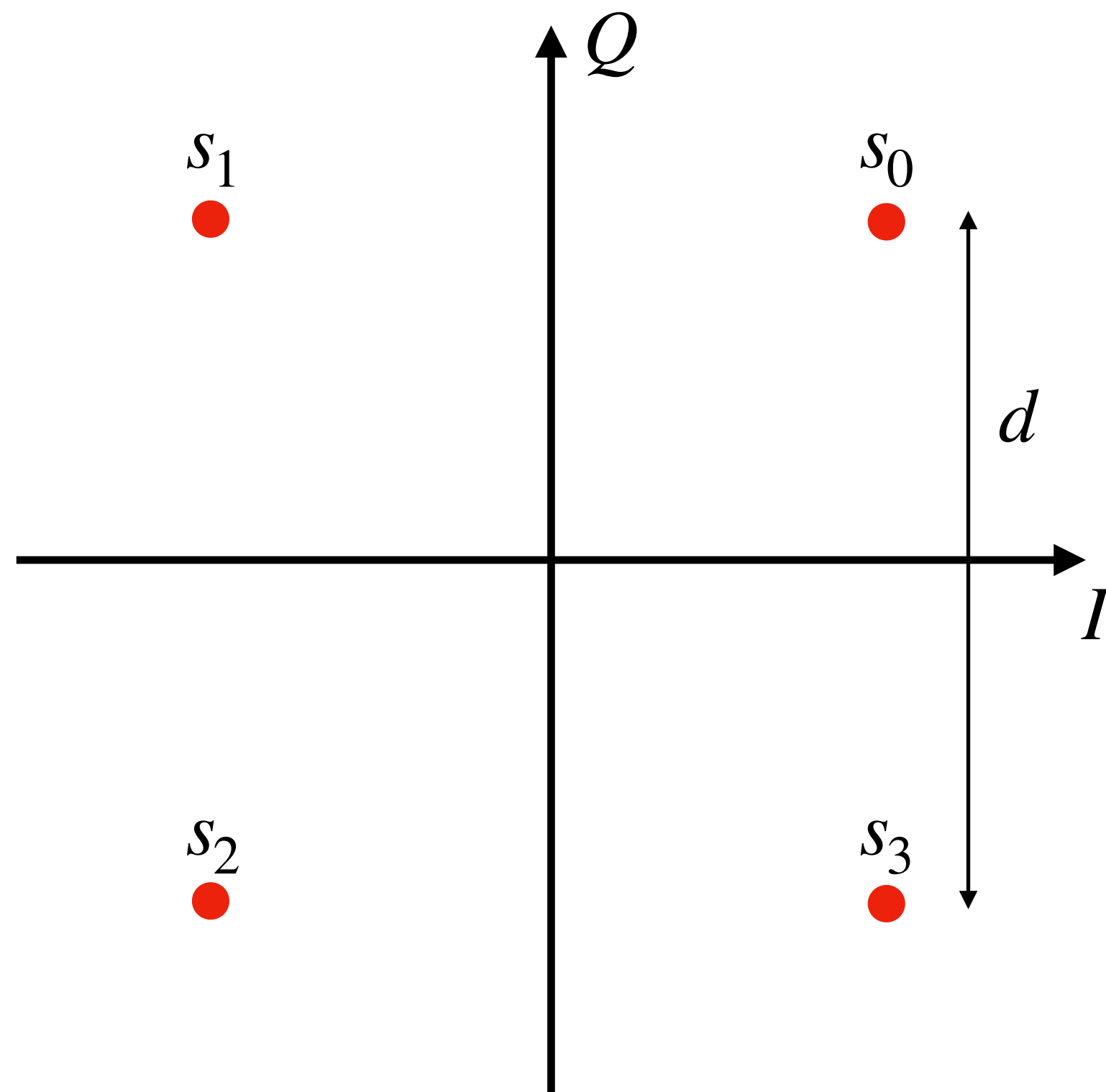
$$\begin{aligned} P_{e|s_0} &= P \left[n_I + d/2 < 0 \text{ or } n_Q + d/2 < 0 \right] \\ &= P \left[n_I + d/2 < 0 \right] + P \left[n_Q + d/2 < 0 \right] - P \left[n_I + d/2 < 0 \text{ and } n_Q + d/2 < 0 \right] \end{aligned}$$

$$P \left[n_I + d/2 < 0 \right] = P \left[n_Q + d/2 < 0 \right] = Q \left(\frac{d}{2\sqrt{N_0}} \right)$$

$$P_{e|s_0} = 2Q \left(\frac{d}{2\sqrt{N_0}} \right) - Q^2 \left(\frac{d}{2\sqrt{N_0}} \right)$$

$$P \left[n_I + d/2 < 0 \text{ and } n_Q + d/2 < 0 \right] = Q^2 \left(\frac{d}{2\sqrt{N_0}} \right)$$

Probability of Error for QPSK



Consider the Constellation for QPSK.

(d) What's the total probability of error given that each symbol is equally likely?

$$P_e = \sum_k P_{e|k} P[k]$$

$$P_e = 2Q\left(\frac{d}{2\sqrt{N_0}}\right) - Q^2\left(\frac{d}{2\sqrt{N_0}}\right)$$

The power of coding!

a) Convolutional Coding takes advantage of what to correct errors at the receiver?

Redundancy

b) How many errors can a single parity-check detect?

1 or an odd number, but no information on which bit has an error.

c) Hamming codes use m equations to correct one error in:

$2^m - 1$ bits

c) Data $d = (1,0,1,1)$ is encoded using a Hamming(7,4) code, what is transmitted?

	c ₁	c ₂	c ₃	c ₄	c ₅	c ₆	c ₇
	p ₁	p ₂	d ₁	p ₄	d ₂	d ₃	d ₄
	1	0	1	0	1	0	1
	0	1	1	0	0	1	1
	0	0	0	1	1	1	1

$$p_1 = d_1 \oplus d_2 \oplus d_4 = 0$$

0110011

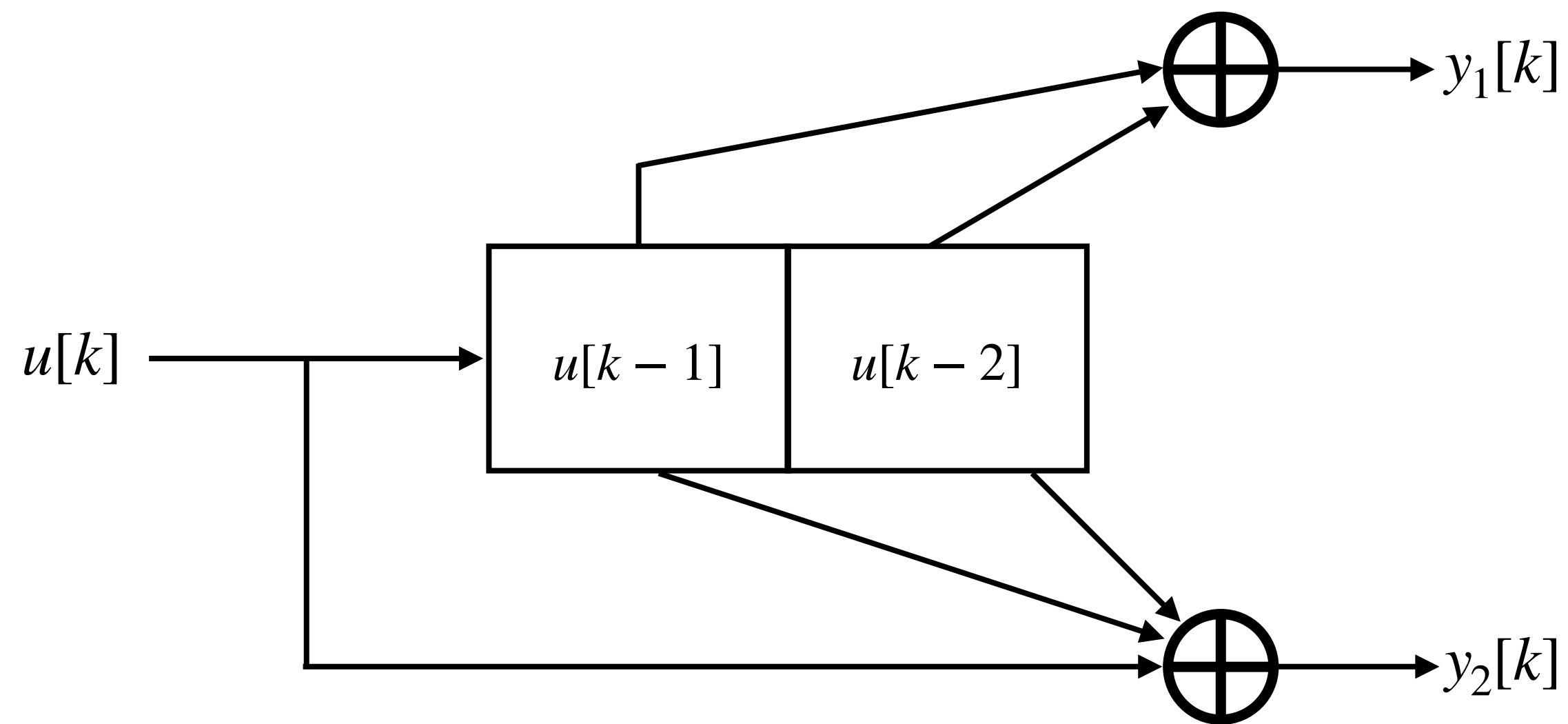
$$p_2 = d_1 \oplus d_3 \oplus d_4 = 1$$

0100011

$$p_4 = d_2 \oplus d_3 \oplus d_4 = 0$$

Error in $p_4 p_2 p_1$

Convolutional Codes



Consider the Shift-register Implementation of a Convolutional Code

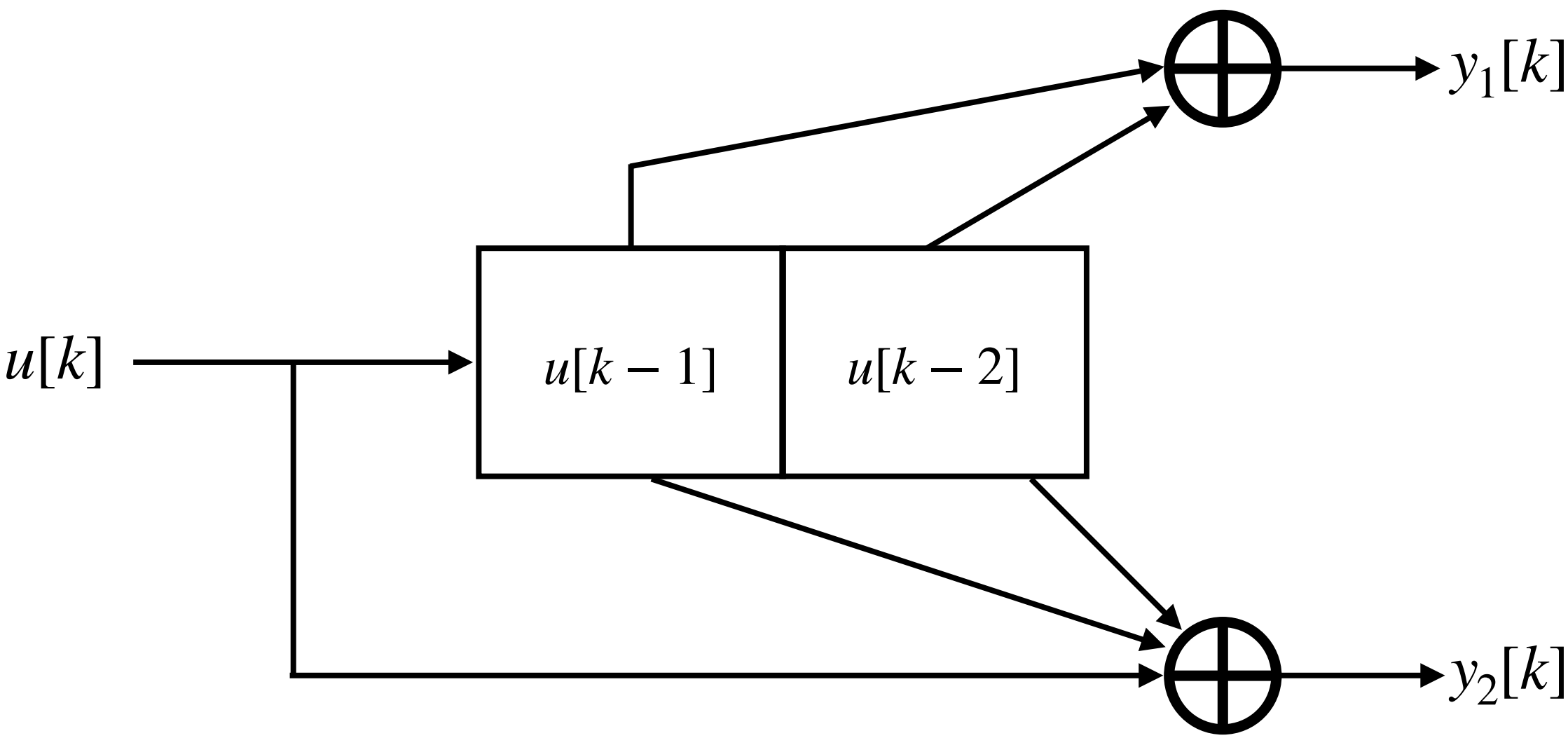
(a) Define the Coding Generator matrix $G(D)$.

$$g_{11}(D) = 011_{(2)}$$

$$g_{12}(D) = 111_{(2)}$$

$$\begin{bmatrix} u[k] & s[k] \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} y_1[k] & y_2[k] \end{bmatrix}$$

Convolutional Codes

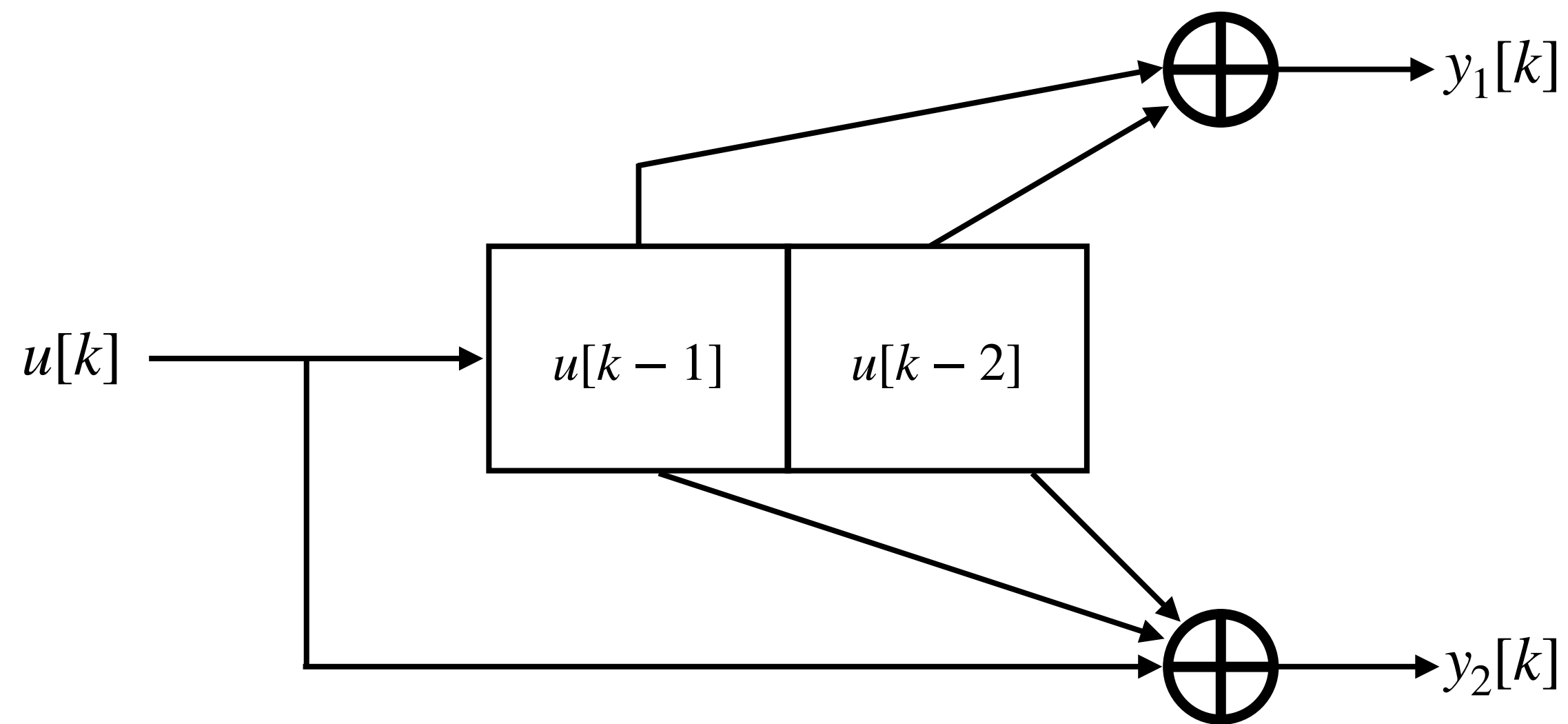


Consider the Shift-register Implementation of a Convolutional Code

(b) Fill in the state table below

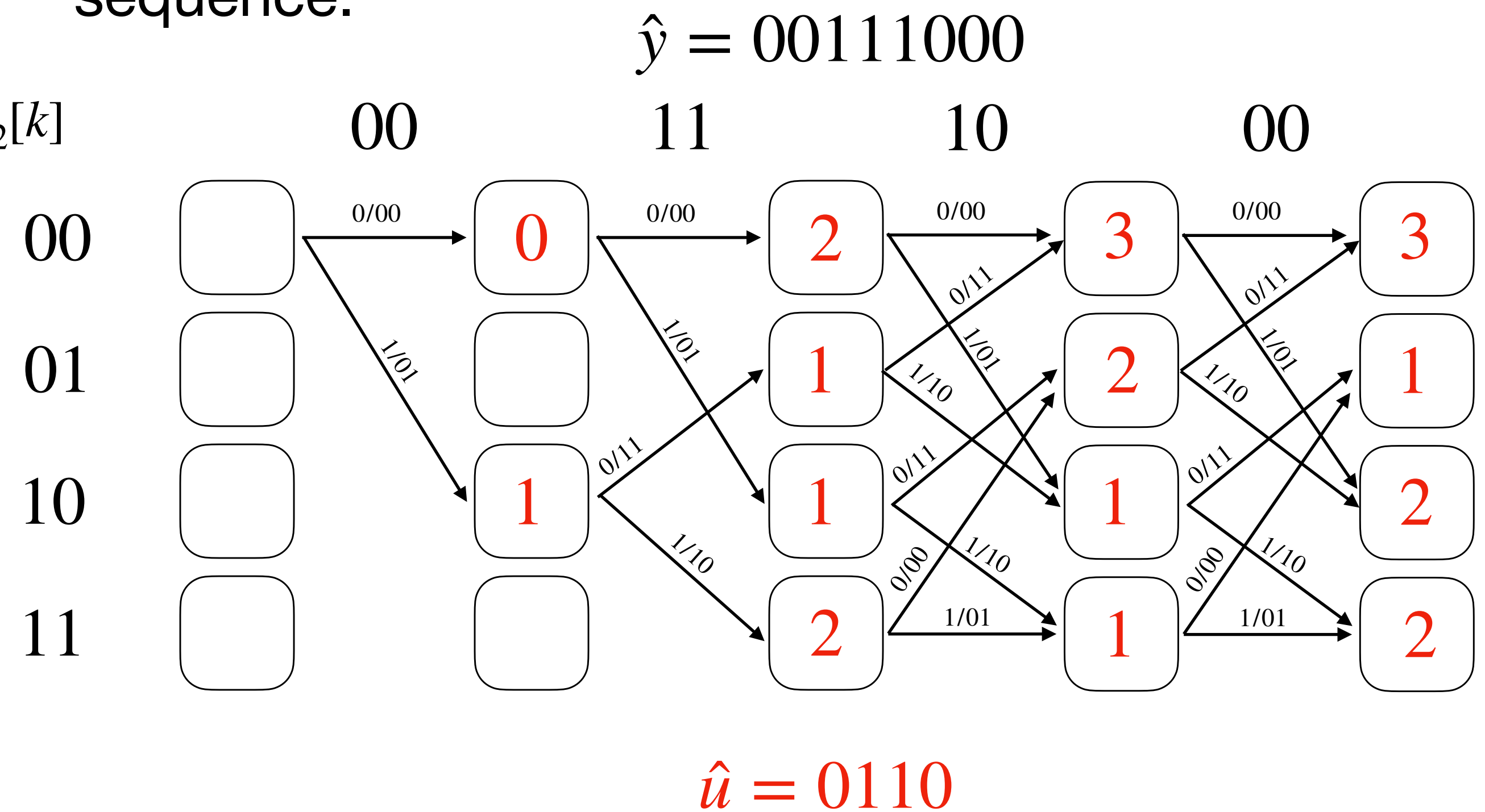
k	$u[k]$	$s[k]$	$y_1[k]$	$y_2[k]$
0	0	(0,0)	0	0
1	1	(0,0)	0	1
2	1	(1,0)	1	0
3	0	(1,1)	0	0
4	1	(0,1)	1	0
5	0	(1,0)	1	1

Convolutional Codes



Consider the Shift-register Implementation of a Convolutional Code

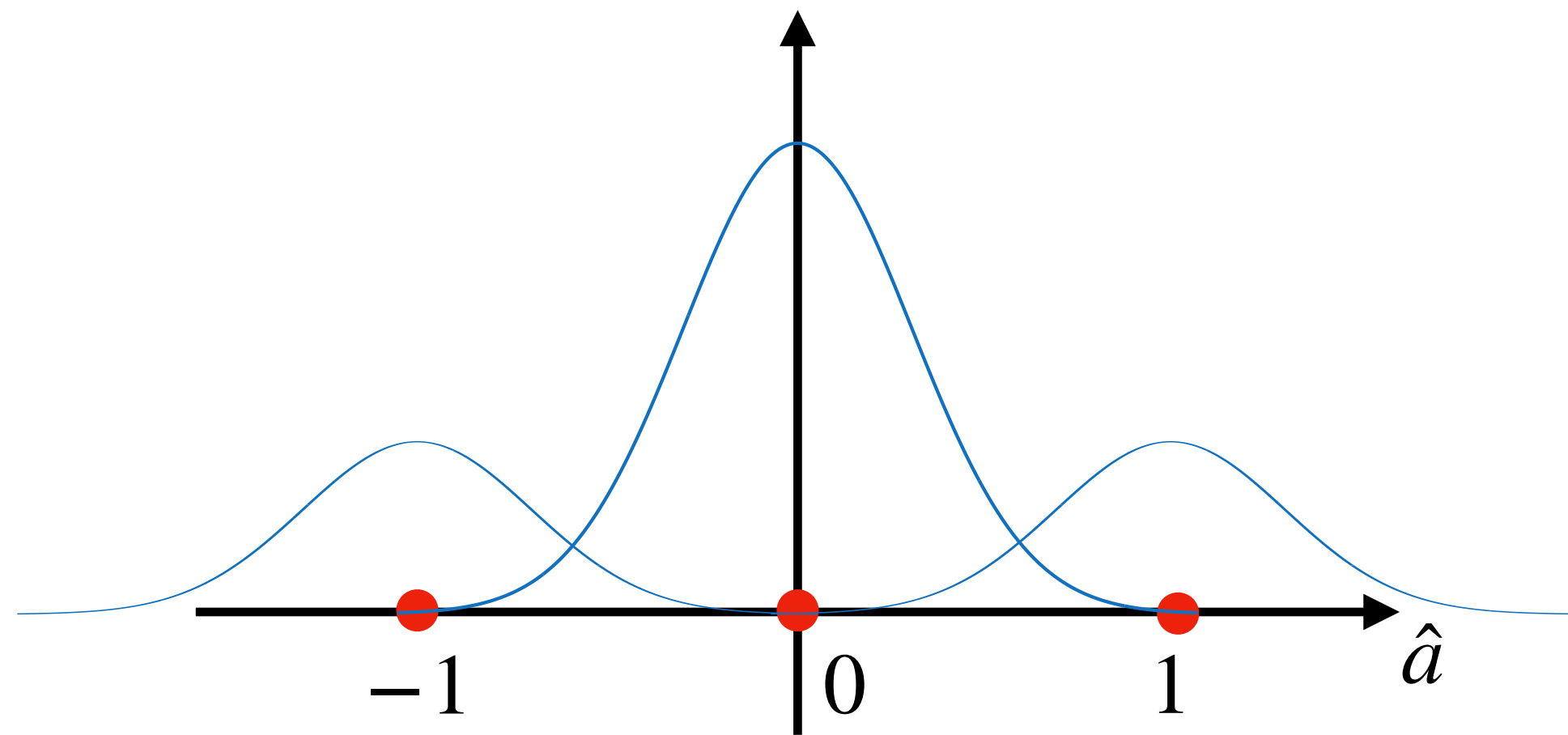
(c) You received $\hat{y}$, draw a Trellis diagram and use the Viterbi Algorithm to estimate the transmitted bit sequence.



MAP vs. ML Estimation

Consider PAM3 where the likelihood of each symbol varies with $P[0] = 1/2$, $P[-1] = P[1] = 1/4$. The received signal has Gaussian Noise with variance N_0 .

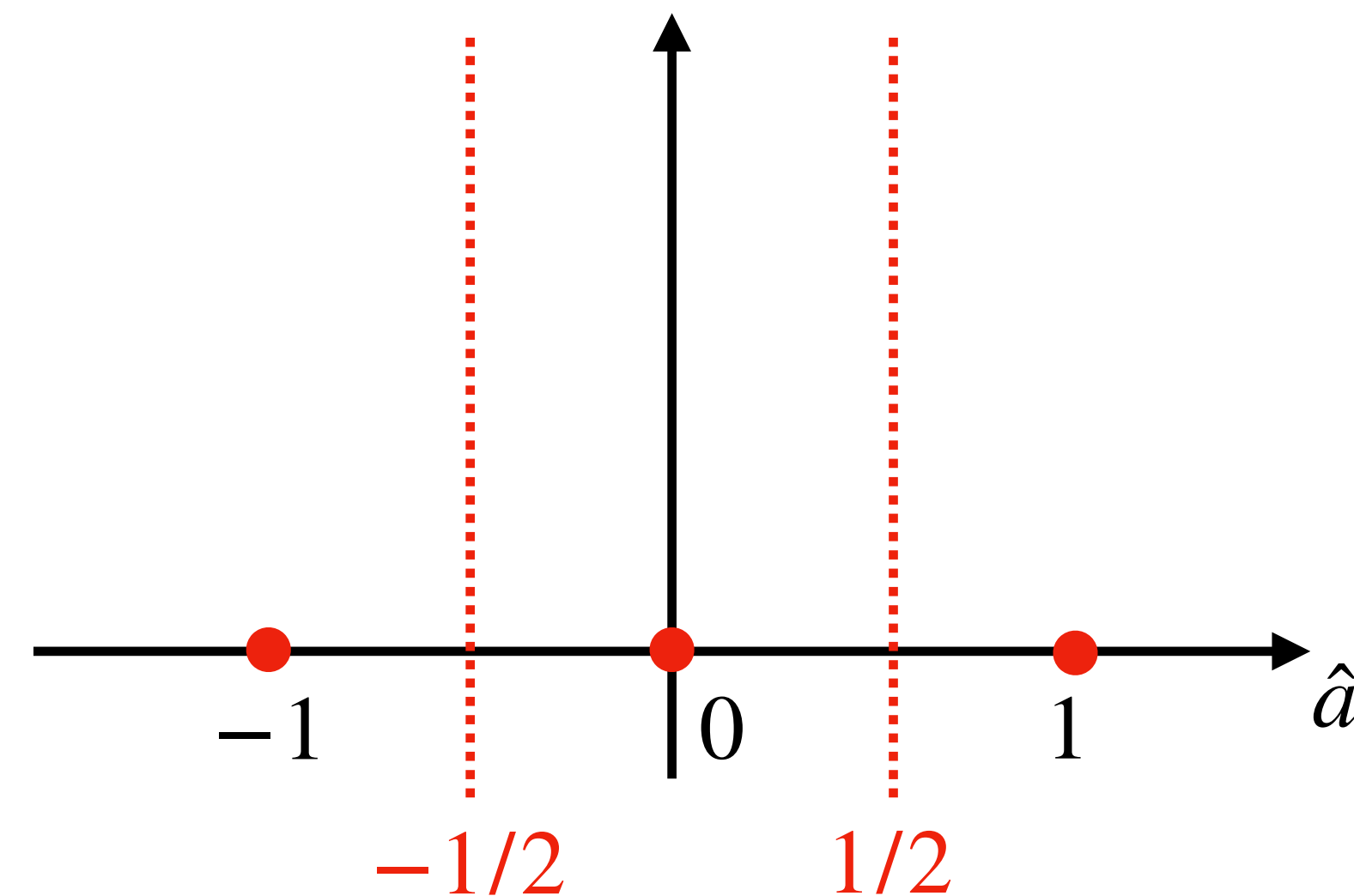
(a) Using Maximum Likelihood (ML) Estimation, draw the decision boundaries and compute the probability of error.



$$P_e = P[0]P_{e|0} + P[1]P_{e|1} + P[-1]P_{e|-1}$$

$$P_{e|0} = P[\hat{a} > 1/2] + P[\hat{a} < -1/2] = 2Q\left(\frac{1}{2\sqrt{N_0}}\right)$$

$$P_{e|1} = P_{e|-1} = Q\left(\frac{1}{2\sqrt{N_0}}\right)$$

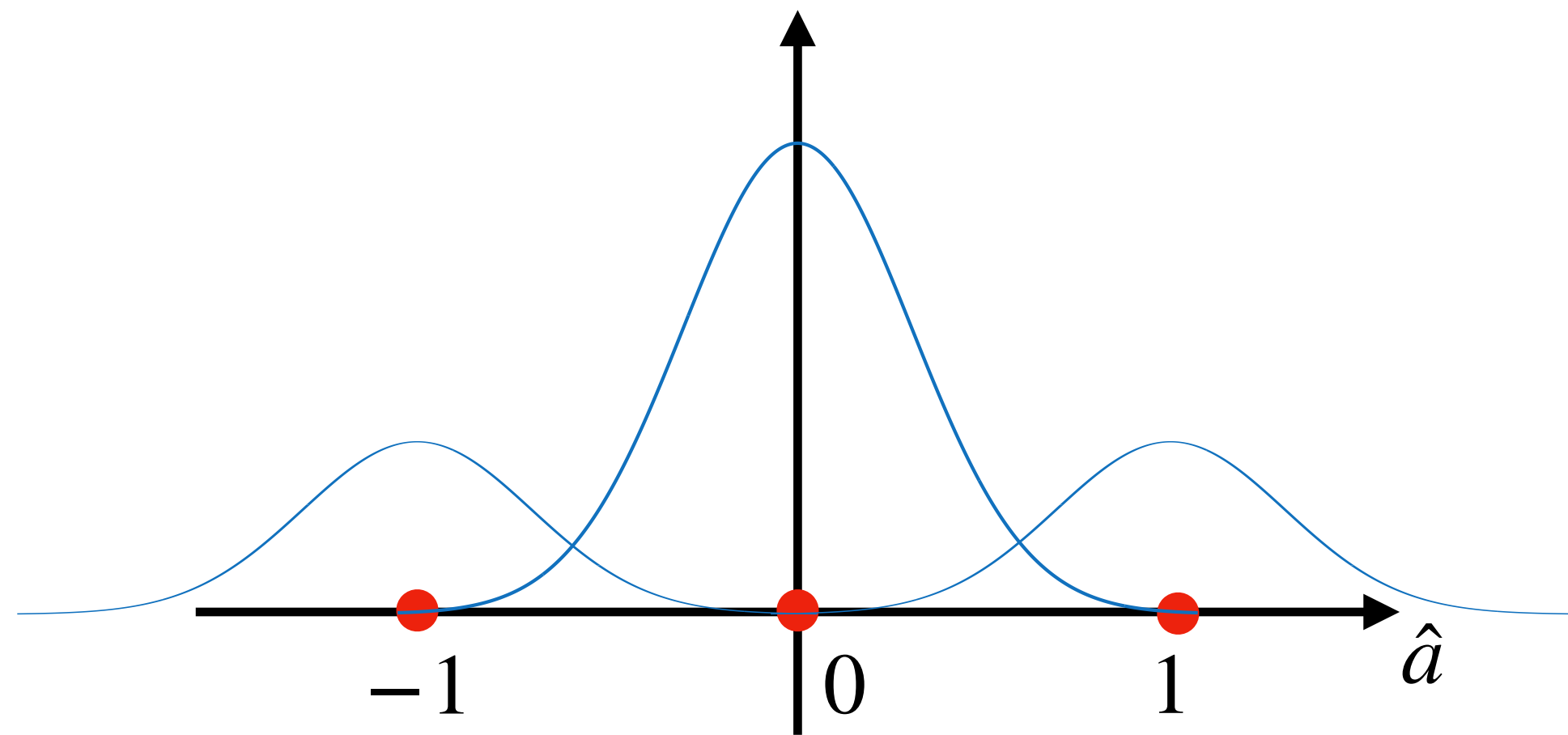


$$P_e = \frac{3}{2}Q\left(\frac{1}{2\sqrt{N_0}}\right)$$

MAP vs. ML Estimation

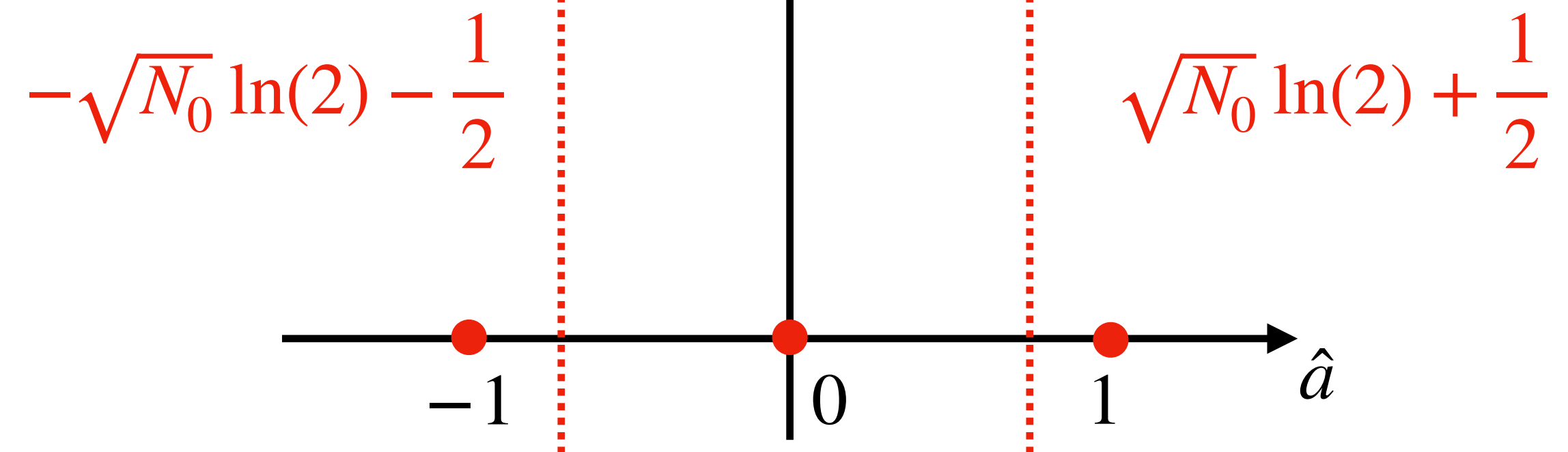
Consider PAM3 where the likelihood of each symbol varies with $P[0] = 1/2$, $P[-1] = P[1] = 1/4$. The received signal has Gaussian Noise with variance N_0 .

(b) Using MAP Estimation, draw the decision boundaries. How are they different from ML?

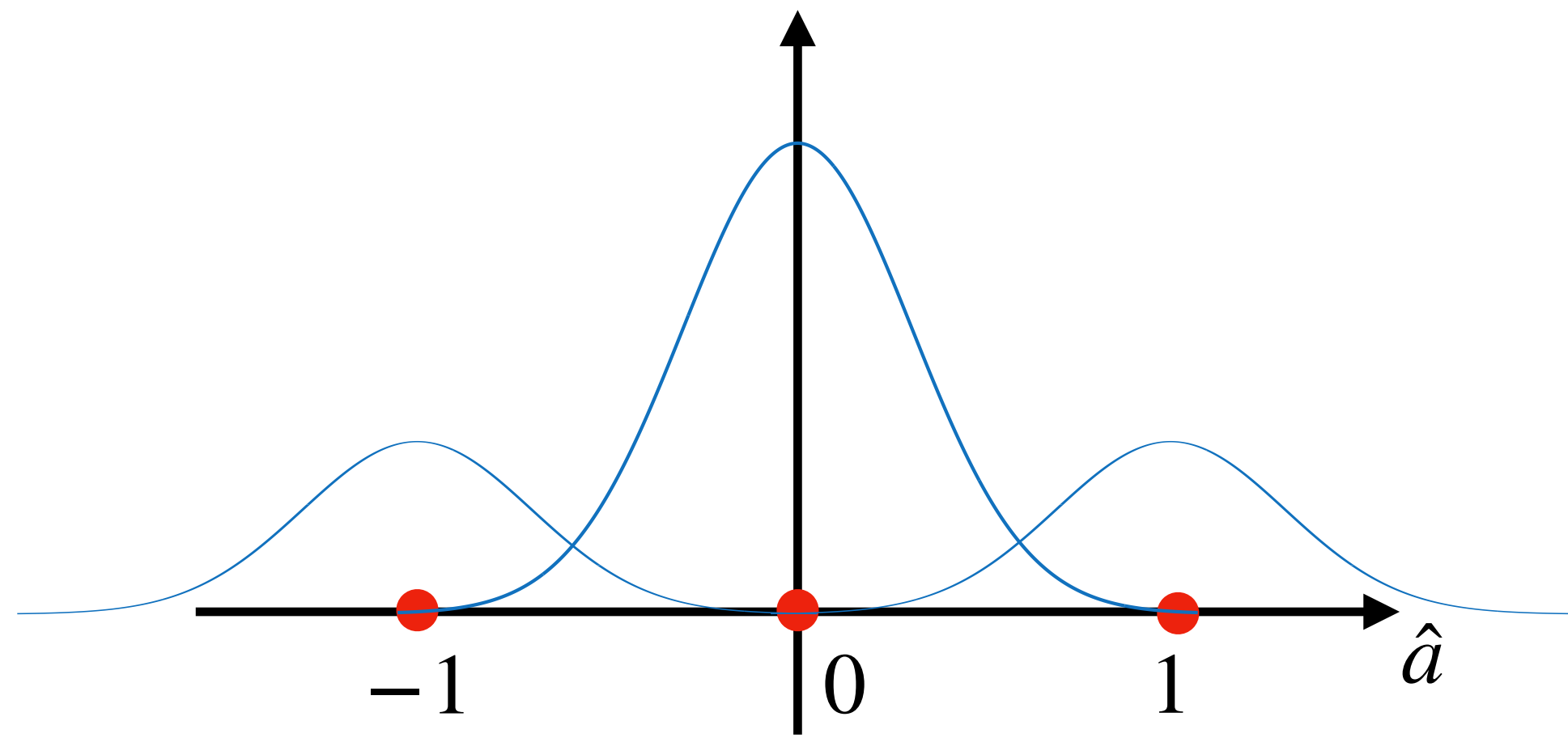


$$P[0] \frac{1}{\sqrt{2\pi N_0}} e^{-\frac{\hat{a}^2}{2N_0}} = P[1] \frac{1}{\sqrt{2\pi N_0}} e^{-\frac{(\hat{a}-1)^2}{2N_0}}$$

$$\hat{a} = \sqrt{N_0} \ln(2) + \frac{1}{2}$$



MAP vs. ML Estimation



Consider PAM3 where the likelihood of each symbol varies with $P[0] = 1/2$, $P[-1] = P[1] = 1/4$. The received signal has Gaussian Noise with variance N_0 .

(c) What's the probability of error using the MAP estimator?

$$P_e = P[0]P_{e|0} + P[1]P_{e|1} + P[-1]P_{e|-1}$$

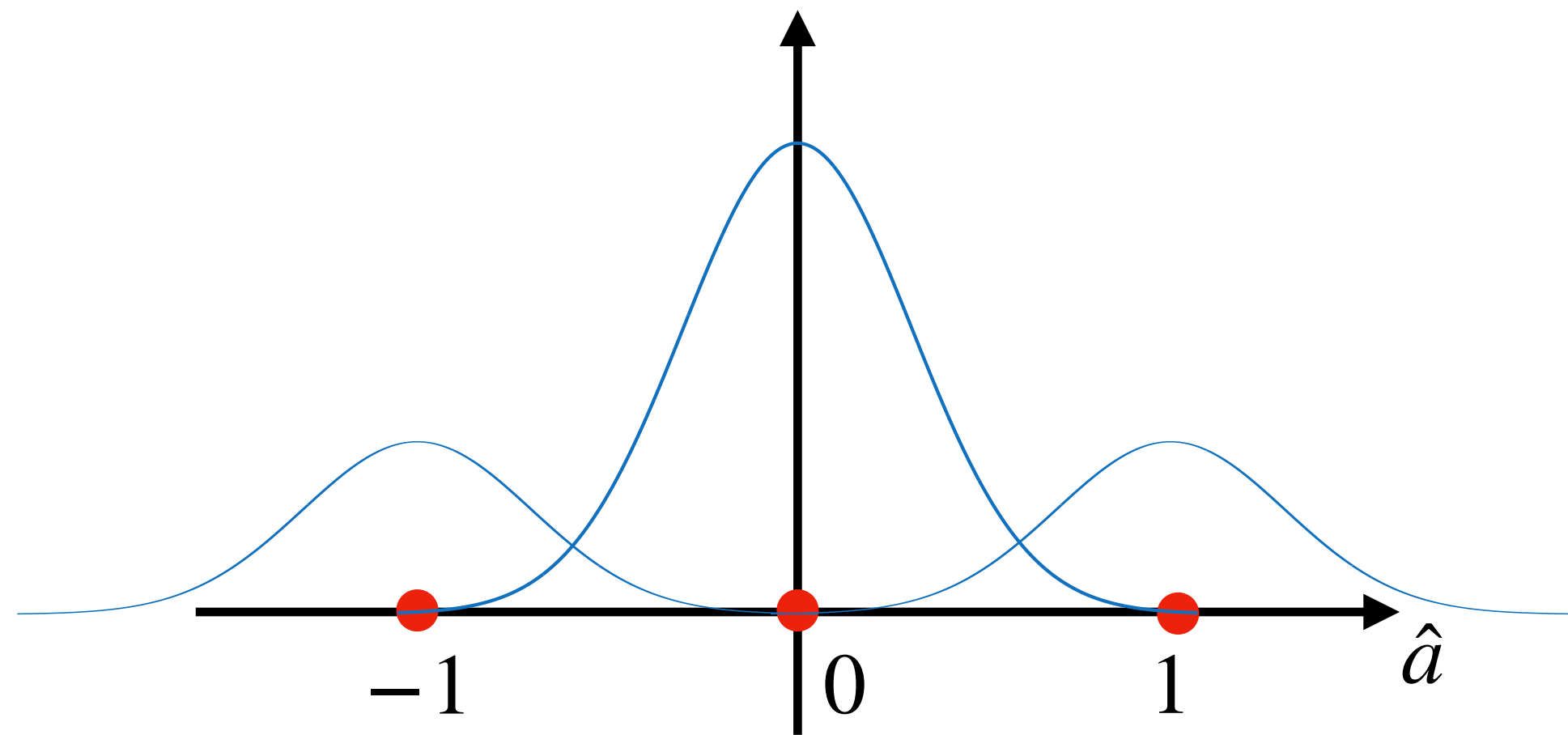
$$P_{e|0} = P[\hat{a} > \beta] + P[\hat{a} < -\beta] = 2Q\left(\frac{\beta}{\sqrt{N_0}}\right)$$

$$P_{e|1} = P_{e|-1} = Q\left(\frac{1 - \beta}{\sqrt{N_0}}\right)$$

$$\beta = \sqrt{N_0} \ln(2) + \frac{1}{2}$$

$$P_e = Q\left(\frac{\beta}{\sqrt{N_0}}\right) + \frac{1}{2}Q\left(\frac{1 - \beta}{\sqrt{N_0}}\right)$$

MAP vs. ML Estimation



Consider PAM3 where the likelihood of each symbol varies with $P[0] = 1/2$, $P[-1] = P[1] = 1/4$. The received signal has Gaussian Noise with variance N_0 .

(d) $\hat{a} = -0.6$, use a soft decision to determine the received symbol assuming $N_0 = 1$

$$P[H_i | \hat{a}] = \frac{p(\hat{a} | i)P[i]}{p(\hat{a})} = \frac{p(\hat{a} | i)P[i]}{\sum_{j=0}^{M-1} p(\hat{a} | j)p(j)}$$

Power Spectrum of Line Codes

$x[k]$ is a bit sequence that is Bernoulli distributed with $P[x_k = 1] = p$ and $P[x_k = 0] = 1 - p$. Find the autocorrelation of $x[k]$. What type of signaling is this?

$$x[k] \sim \text{Bernoulli}\{0,1\} \rightarrow P[x = 1] = p, P[x = 0] = 1 - p$$

$$\mathbb{E}[x_k] = (1)P(x_k = 1) + (0)P(x_k = 0) = p$$

$$\mathbb{E}[x_k^2] = (1)^2P(x_k = 1) + (0)^2P(x_k = 0) = p$$

$$\mathbb{E}[x_k x_{k+n}] = \mathbb{E}[x_k] \mathbb{E}[x_{k+n}] = p^2$$

$$R_x[n] = p^2(1 + (1/p - 1)\delta[n])$$

This is On-Off Keying but the probabilities of '1' and '0' are not equal.

Power Spectrum of Line Codes

The bit sequence $x[k]$ transmitted at a rate of T_b using a rectangular pulse $p(t) = \Pi(2t/T_b)$. Find $P(f)$ and the Power Spectrum of the transmitted signal.

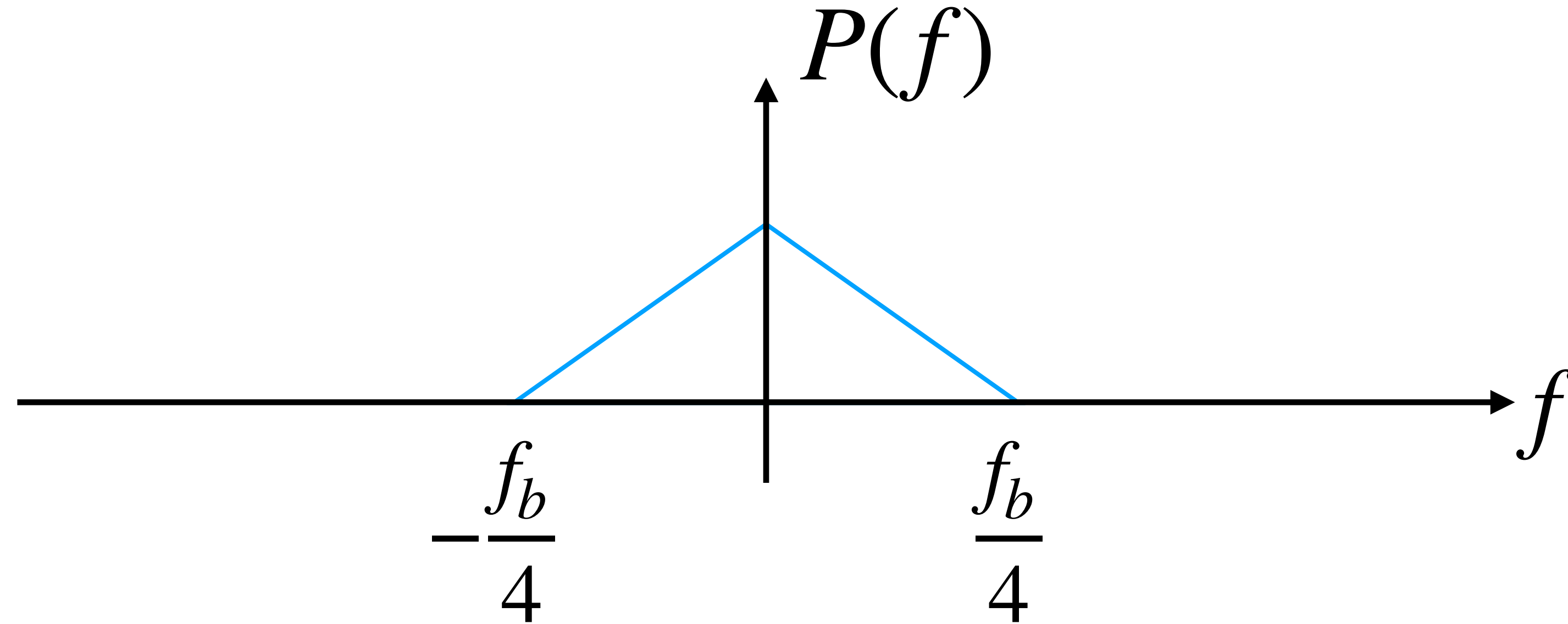
$$p(t) = \Pi(2t/T_b) \rightarrow P(f) = \frac{T_b}{2} \text{sinc}(fT_b/2)$$

$$|P(f)|^2 = \frac{T_b^2}{4} \text{sinc}^2(fT_b/2)$$

$$\text{DTFT}\{R_x[n]\} = p^2(\delta(f) + (1/p - 1)) \rightarrow S_x(f) = \frac{p(1-p)}{T_b} + \frac{p^2}{T_b} \sum_n \delta(f - n/T_b)$$

$$S_y(f) = |P(f)|^2 \left(\frac{1}{T_b} \sum_n R_x[n] e^{-j2\pi f n T_b} \right) = |P(f)|^2 S_x(f)$$

Nyquist Pulses



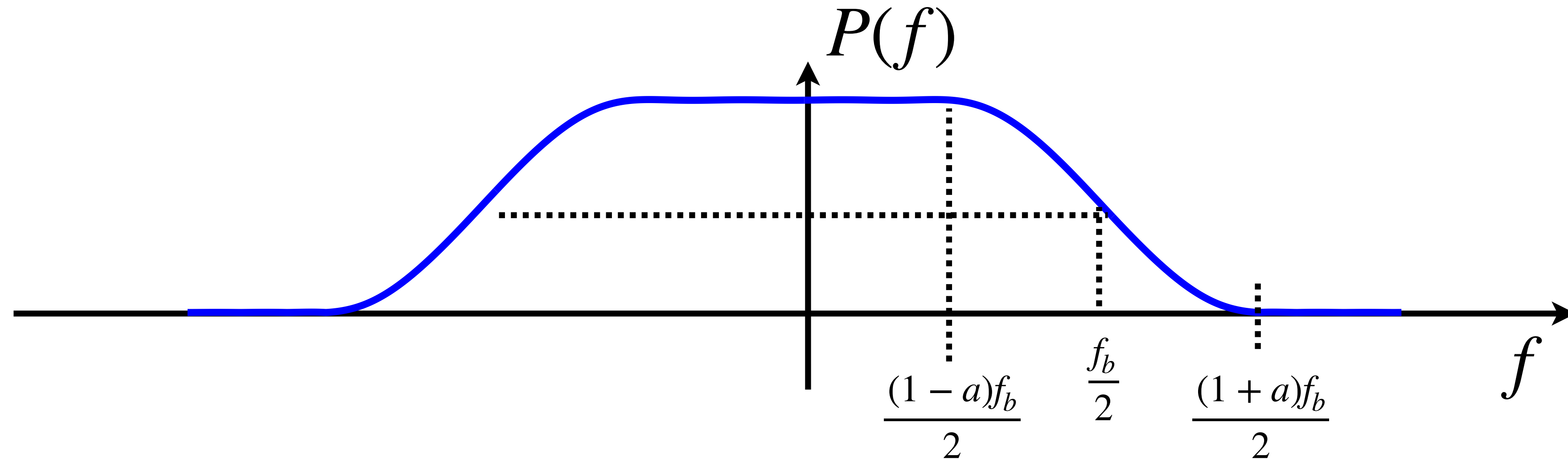
Is this a Nyquist Pulse for Baud Rate f_b ?

$$P(f) \sim \Pi(4f/f_b) * \Pi(4f/f_b)$$

$$p(t) \sim \text{sinc}^2(tf_b/4)$$

$$p(t) \neq \begin{cases} 1 & t = 0 \\ 0 & t = \pm kT_b \end{cases}$$

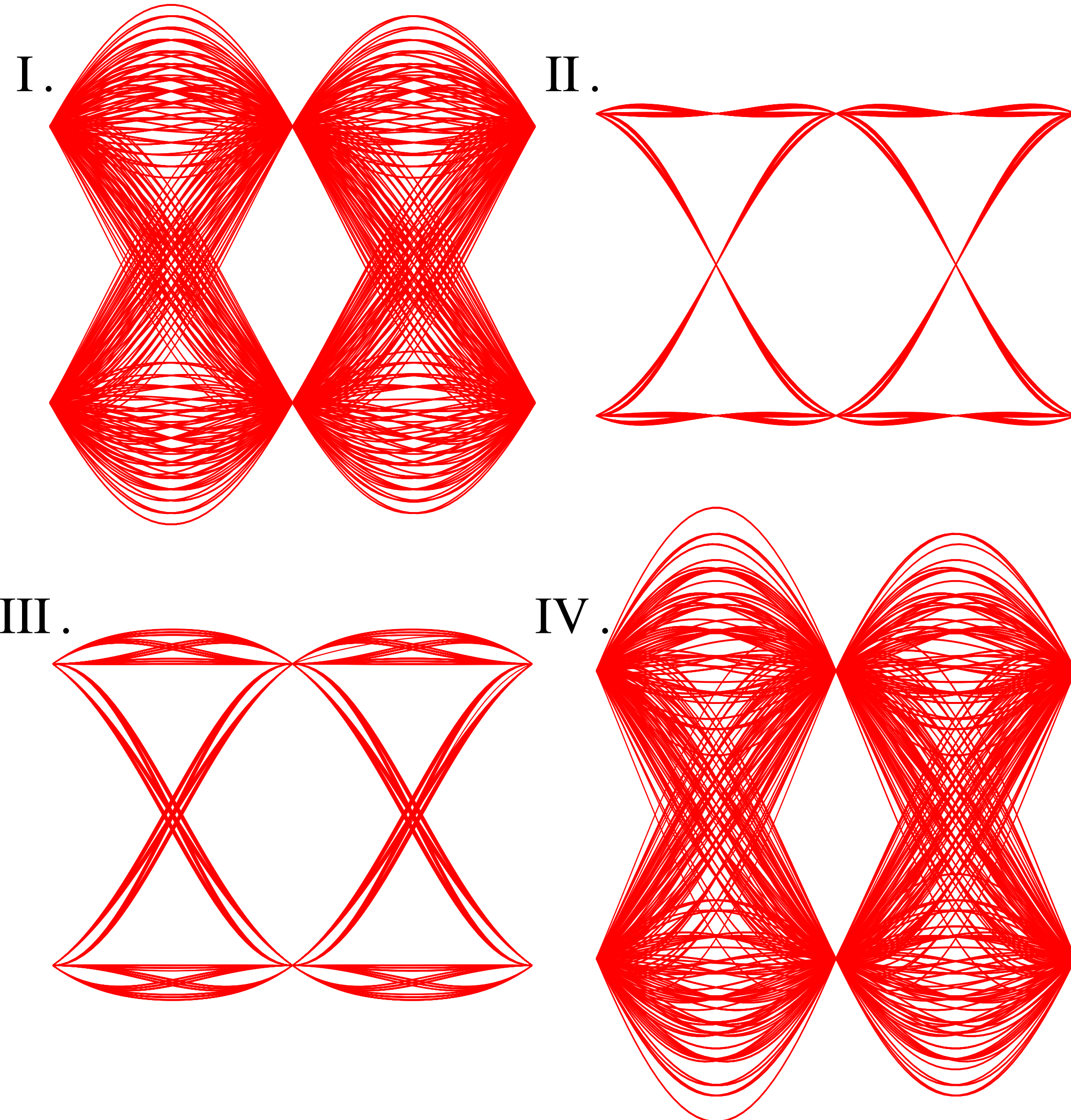
Nyquist Pulses



Is this a Nyquist Pulse for Baud Rate f_b ? What's the excess Bandwidth?

Yes, this is a Raised cosine pulse with excess Bandwidth af_b

Eye Diagrams



Consider the four eye diagrams. Each uses a Raised Cosine Pulse with β values 0, 0.25, 0.75, 1.

(a) How many symbols do the eye diagrams show?

2

(b) Match the β values to the corresponding eye diagram.

I. $\beta = 0.25$, II. $\beta = 1$, III. $\beta = 0.75$, IV. $\beta = 0$

(c) With respect to f_b , how much excess bandwidth does each diagram express?

βf_b

(d) Which do you prefer to minimize timing errors?

Larger β

Abs. Value Of Polar RZ Contains Timing Info!